Retail Sales Analytics Report

Dataset Range: October – December 2020
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1. Introduction

This project analyzes retail sales data from October to December 2020. The goal is to uncover purchasing trends, identify high-performing segments, and provide actionable insights for business decision-making.

2. Methodology

- **Data Cleaning:** Standardized date formats, handled missing values, and ensured consistent category labels.
- **Exploratory Analysis:** Aggregated sales by month, customer type, state, and time of day.
- **Visualization:** Created charts using Python (Pandas, Matplotlib, Seaborn) to highlight trends and anomalies.

3. Key Visuals & Insights

- **Dual Axis Chart (Sales vs. Quantity):** December showed the highest revenue and volume, outperforming October and November.
- **Boxplot (Sales by Customer Type):** Senior customers consistently spent more than other groups.
- **Daily Trend Line:** Evening shopping hours generated peak sales, especially in December.
- **State-wise Sales Chart:** Certain states contributed disproportionately to total revenue, highlighting regional opportunities.

4. Findings

- **Seasonality:** December sales were significantly higher, suggesting holiday demand.
- **Customer Segments:** Seniors were the most valuable segment, especially during evening hours.
- **Regional Performance:** A few states drove the majority of sales, indicating where marketing efforts should be concentrated.
- **Category Trends:** Some product categories consistently underperformed, signaling potential areas for optimization.

5. Challenges & Learnings

- Handling inconsistent date formats required careful preprocessing.
- Dual-axis scaling needed clear labeling to avoid misinterpretation.
- Selecting only the most impactful visuals improved storytelling and recruiter readability.

6. Conclusion

Retailers can replicate December's strategies to boost sales in other months. Focusing on senior customers and evening shopping promotions may yield higher returns. Regional targeting can further optimize marketing spend.